

Chapter 6 - TED Video

The TED chip reproduces the picture model of the C16 and Plus/4. It sits on compositor layer 12, between VGA (layer 10) and ANTIC (layer 13). TED's design folds video and audio into one chip; this chapter covers the video half. The audio half is described in Chapter 16.

6.1 What TED can show

Item	Value
Display area	320 × 200 pixels
Character grid	40 × 25 cells of 8 × 8
Border	32 left, 32 right, 35 top, 37 bottom
Total frame	384 × 272 pixels
Colours	121 unique (16 hues × 8 luminances minus duplicated blacks)
Modes	Text, bitmap, multicolour bitmap, extended-colour text

In multicolour mode each pixel is two screen pixels wide, so the effective resolution drops to 160 × 200 with four colours per cell instead of two.

6.2 BASIC keywords

The BASIC keyword TED introduces every TED subcommand. The recognised forms are:

Form	Effect
TED ON	Enable the chip (sets TED_V_ENABLE bit 0).
TED OFF	Disable the chip.
TED MODE n	0 = text, 1 = bitmap, 2 = multicolour bitmap. Sets the BMM and MCM bits and forces DEN on.
TED COLOR bg [, $border$]	Set background colour 0 and (optionally) the border colour.
TED CHAR $addr$	Set the character/bitmap base register.
TED VIDEO $addr$	Set the video matrix base register.
TED CLS	Fill the selected video matrix with the space character (\$20).
TED SCROLL dx, dy	Write the X and Y fine-scroll fields of CTRL2 and CTRL1 (each 0-7).

For anything beyond the simplest setup, use POKE and POKE8 on the registers and TED VRAM. This is the normal TED programming style, because the useful picture data is just bytes in the matrix, colour RAM, character set, and bitmap area.

This first program enables TED text mode and writes a coloured banner directly into the video matrix:

```

10 TED ON
20 POKE &H000F0F20, &H18      : REM DEN + RSEL
30 POKE &H000F0F24, &H08      : REM CSEL
40 POKE &H000F0F30, &H06      : REM blue background
50 POKE &H000F0F40, &H5E      : REM bright dark-blue border
60 FOR I=0 TO 999
70 POKE8 &H000F3000+I, 32
80 POKE8 &H000F3400+I, &H71
90 NEXT I
100 T$="TED RAINBOW"
110 FOR I=1 TO LEN(T$)
120 A=&H000F3000+12*40+14+I
130 C=&H000F3400+12*40+14+I
140 POKE8 A, ASC(MID$(T$,I,1))
150 POKE8 C, &H10+I
160 NEXT I

```

You should see a TED text screen with a dark-blue border and a multi-coloured message near the centre.

6.3 The register block

TED's video registers live in the small block \$F0F20-\$F0F6B. The chip's audio registers sit at \$F0F00-\$F0F05 (Chapter 16).

Address	Name	Purpose
\$F0F20	TED_V_CTRL1	Control 1 (ECM, BMM, DEN, RSEL, YSCROLL).
\$F0F24	TED_V_CTRL2	Control 2 (RES, MCM, CSEL, XSCROLL).
\$F0F28	TED_V_CHAR_BASE	Character generator base (high nibble) and bitmap base (low nibble), in 1 KB steps.
\$F0F2C	TED_V_VIDEO0_BASE	Video matrix base, in 1 KB steps (bits 3-7).
\$F0F30	TED_V_BG_COLOR0	Background colour 0.
\$F0F34	TED_V_BG_COLOR1	Background colour 1 (multicolour).
\$F0F38	TED_V_BG_COLOR2	Background colour 2 (multicolour).
\$F0F3C	TED_V_BG_COLOR3	Background colour 3 (multicolour).
\$F0F40	TED_V_BORDER	Border colour.
\$F0F44	TED_V_CURSOR_HI	Cursor position, high byte.
\$F0F48	TED_V_CURSOR_LO	Cursor position, low byte.
\$F0F4C	TED_V_CURSOR_CLR	Cursor colour.
\$F0F50	TED_V_RASTER_LO	Current raster line, low byte (read-only).
\$F0F54	TED_V_RASTER_HI	Raster bit 8 in bit 0 (read-only).
\$F0F58	TED_V_ENABLE	Bit 0 = video enable.
\$F0F5C	TED_V_STATUS	Bit 0 = VBlank active (read-only).
\$F0F60	TED_V_RASTER_CMP_LO	Raster compare line, low byte.
\$F0F64	TED_V_RASTER_CMP_HI	Raster compare line bit 8 in bit 0.
\$F0F68	TED_V_RASTER_STATUS	Bit 7 = compare pending; write \$80 to clear.

Every register is a 32-bit word at a 4-byte-aligned address; only the low byte of each is meaningful.

6.3.1 CTRL1 bits

Bit	Name	Meaning
6	ECM	Extended colour mode.
5	BMM	Bitmap mode.
4	DEN	Display enable. Must be set for any picture.
3	RSEL	0 = 24 rows, 1 = 25 rows.
0-2	YSCROLL	Vertical fine-scroll, 0-7 raster lines.

6.3.2 CTRL2 bits

Bit	Name	Meaning
5	RES	Reset.
4	MCM	Multicolour mode.
3	CSEL	0 = 38 columns, 1 = 40 columns.
0-2	XSCROLL	Horizontal fine-scroll, 0-7 pixels.

The picture mode is the combination of BMM, MCM, and ECM:

BMM	MCM	ECM	Picture
0	0	0	Standard text.
0	1	0	Multicolour text.
0	0	1	Extended-colour text (8 distinct background colours from the top three bits of the character code).
1	0	0	Standard bitmap.
1	1	0	Multicolour bitmap (160 × 200).

6.4 The colour byte

Every colour register and every byte of colour RAM holds an 8-bit **colour byte** with this layout:

```
bit 7 6 5 4 3 2 1 0
    0 L L L H H H H
      |-luminance| |---hue---|
```

L is the luminance (0-7); H is the hue (0-15). The hue table is:

Hue	Name	Hue	Name
0	Black	8	Orange
1	White	9	Brown
2	Red	10	Yellow-green
3	Cyan	11	Pink

Hue	Name	Hue	Name
4	Purple	12	Blue-green
5	Green	13	Light blue
6	Blue	14	Dark blue
7	Yellow	15	Light green

Hue 0 (black) is the same colour at every luminance, which is why TED has 121 unique colours rather than 128 (16×8).

6.5 The VRAM region

TED owns 16 KB of private VRAM beginning at \$F3000. This region holds three things, all of which can be relocated through the TED_V_VIDEO0_BASE and TED_V_CHAR_BASE registers:

Block	Default offset (from \$F3000)	Size
Video matrix	\$0000	1024 B
Colour RAM	\$0400	1024 B
Character set	\$0800	2048 B

The video matrix is a 40×25 array of character codes. The colour RAM is a 40×25 array of colour bytes, one per cell. The character set is 256 entries of 8 bytes each; bit 7 of every byte is the leftmost pixel.

TED_V_VIDEO0_BASE selects the video matrix base in 1 KB steps, through bits 3-7 of its low byte. TED_V_CHAR_BASE packs two selectors into one byte: bits 4-7 choose the character set, and bits 0-3 choose the bitmap base when in bitmap mode. Both selectors are also in 1 KB steps. If a selected base would make the needed data run beyond the 16 KB TED VRAM window, the chip falls back to the default base for that data type.

6.6 Text mode data

Standard text mode uses:

- video matrix byte: character code
- colour RAM byte: foreground colour byte
- background colour: TED_V_BG_COLOR0
- character bitmap: eight bytes per character, MSB first

For cell (x, y) , where x is 0-39 and y is 0-24:

```
matrix address = $F3000 + matrixBase + y * 40 + x
colour address = $F3000 + matrixBase + 1024 + y * 40 + x
glyph address  = $F3000 + charBase + charCode * 8 + row
```

When bit 7 of a colour-RAM byte is set, the glyph is drawn in reverse video: set pixels become background and clear pixels become foreground. The displayed colour still uses the low seven bits of the colour byte.

6.6.1 Extended-colour text

Extended-colour text sets ECM in TED_V_CTRL1. The top two bits of each character code select one of TED_V_BG_COLOR0-3; the low six bits select the glyph. Colour RAM remains the foreground colour.

This program makes four wide background panels using the same blank glyph and four different background registers:

```
10 TED ON
20 POKE &H000F0F20, &H58      : REM ECM + DEN + RSEL
30 POKE &H000F0F24, &H08      : REM 40 columns
40 POKE &H000F0F30, &H72      : REM red
50 POKE &H000F0F34, &H75      : REM green
60 POKE &H000F0F38, &H76      : REM blue
70 POKE &H000F0F3C, &H77      : REM yellow
80 POKE &H000F0F40, &H71      : REM white border
90 FOR I=0 TO 999
100 K=1+64*INT((I MOD 40)/10)
110 POKE8 &H000F3000+I, K
120 POKE8 &H000F3400+I, &H71
130 NEXT I
```

The character code is 1, 65, 129, or 193. In extended-colour mode all four codes use glyph 1, while their top two bits select the four background colours.

6.6.2 Multicolour text

Multicolour text clears BMM and sets MCM. The character bitmap is read two bits at a time, so each source pixel is two screen pixels wide:

Bits	Colour source
00	TED_V_BG_COLOR0
01	TED_V_BG_COLOR1
10	TED_V_BG_COLOR2
11	Colour RAM for the cell

The same addressing as standard text mode is used. The trade-off is horizontal detail: an 8-pixel character row becomes four double-wide colour cells.

This program builds one four-colour glyph and fills the screen with it:

```
10 TED ON
20 POKE &H000F0F20, &H18      : REM DEN + RSEL
30 POKE &H000F0F24, &H18      : REM MCM + CSEL
40 POKE &H000F0F30, &H06      : REM background blue
50 POKE &H000F0F34, &H72      : REM multicolour 1 red
60 POKE &H000F0F38, &H75      : REM multicolour 2 green
70 POKE &H000F0F40, &H71
80 FOR R=0 TO 7
90 POKE8 &H000F3800+2*8+R,&H1B
100 NEXT R
110 FOR I=0 TO 999
120 POKE8 &H000F3000+I,2
130 POKE8 &H000F3400+I,&H77
140 NEXT I
```

The glyph byte \$1B is the bit pattern 00,01,10,11, so each character cell repeats all four multicolour text sources.

6.7 Bitmap modes

Bitmap mode sets BMM in TED_V_CTRL1. The bitmap base comes from the low nibble of TED_V_CHAR_BASE:

```
bitmapBase = (TED_V_CHAR_BASE AND 15) * 1024
```

The bitmap contains 200 rows. For pixel row *y*, cell column *cellX*, and row inside the cell *row* = *y* AND 7, the byte address is:

```
$F3000 + bitmapBase + INT(y / 8) * 320 + row * 40 + cellX
```

That is 8000 bytes for a full 320 x 200 bitmap.

6.7.1 High-resolution bitmap

High-resolution bitmap mode clears MCM. Each bitmap bit is one screen pixel. The video matrix byte for the cell supplies the two colours:

Bitmap bit	Colour source
0	low nibble of the matrix byte
1	high nibble of the matrix byte

This example selects a bitmap at TED VRAM offset \$2000 and draws yellow and blue hatch stripes:

```
10 TED ON
20 POKE &H000F0F20, &H38      : REM BMM + DEN + RSEL
30 POKE &H000F0F24, &H08      : REM hires, 40 columns
40 POKE &H000F0F28, &H28      : REM bitmap base $2000
50 POKE &H000F0F40, &H71
60 FOR I=0 TO 999
70 POKE8 &H000F3000+I, &H76    : REM bit 1 yellow, bit 0 blue
80 NEXT I
90 FOR Y=0 TO 199
100 FOR X=0 TO 39
110 A=&H000F5000+INT(Y/8)*320+(Y AND 7)*40+X
120 IF (Y AND 8)=0 THEN POKE8 A,&HAA ELSE POKE8 A,&H55
130 NEXT X
140 NEXT Y
```

6.7.2 Multicolour bitmap

Multicolour bitmap mode sets both BMM and MCM. The bitmap is read as two-bit pairs, and each pair draws a double-wide pixel:

Bits	Colour source
00	TED_V_BG_COLOR0
01	high nibble of the matrix byte
10	low nibble of the matrix byte
11	colour RAM byte for the cell

This example fills the screen with repeating four-colour vertical bars:

```

10 TED ON
20 POKE &H000F0F20, &H38      : REM BMM + DEN + RSEL
30 POKE &H000F0F24, &H18      : REM MCM + CSEL
40 POKE &H000F0F28, &H28      : REM bitmap base $2000
50 POKE &H000F0F30, &H06      : REM background blue
60 FOR I=0 TO 999
70 POKE8 &H000F3000+I, &H75    : REM green and blue nibbles
80 POKE8 &H000F3400+I, &H72    : REM bright red
90 NEXT I
100 FOR Y=0 TO 199
110 FOR X=0 TO 39
120 A=&H000F5000+INT(Y/8)*320+(Y AND 7)*40+X
130 POKE8 A, &H1B              : REM pairs 00,01,10,11
140 NEXT X
150 NEXT Y

```

6.8 Fine scrolling and visible area

YSCROLL delays the visible picture by 0-7 raster lines. XSCROLL delays it by 0-7 pixels. Newly exposed pixels at the top or left remain border/background until the scrolled source coordinates enter the display area.

RSEL and CSEL choose the full text area:

Bit	Value 0	Value 1
RSEL	24 rows, first and last text rows blanked	25 rows
CSEL	38 columns, first and last text columns blanked	40 columns

Try this after the text-banner program above:

```

200 FOR S=0 TO 7
210 TED SCROLL S, S
220 FOR D=1 TO 120
230 NEXT D
240 NEXT S
250 TED SCROLL 0, 0

```

The picture slides down and right in eight small steps, then returns to the normal alignment.

6.9 The cursor

The cursor is a single character cell. Its position is the 11-bit linear offset stored in TED_V_CURSOR_HI/L0 (row * 40 + col), and its colour is in TED_V_CURSOR_CLR. The cursor blinks automatically at one cycle every 30 frames. It is drawn as an underline after the rest of the frame has been rendered, so it appears over text and bitmap pixels in its cell.

This puts the cursor on row 12, column 20, in bright yellow:

```

10 P=12*40+20
20 POKE &H000F0F44, INT(P/256)
30 POKE &H000F0F48, P AND 255
40 POKE &H000F0F4C, &H77

```

6.10 Raster compare and status

TED_V_RASTER_LO and bit 0 of TED_V_RASTER_HI report the current scanline as a 9-bit value. The rendered TED frame is 0-271. Compare values outside the rendered frame are stored, but do not set the pending latch during the current visible frame.

Writing the same 9-bit value into the compare registers arms the raster compare. When the raster reaches the line, TED_V_RASTER_STATUS bit 7 is set and the chip raises its interrupt line. The CPU clears the latch by writing 1 to bit 7 of TED_V_RASTER_STATUS.

TED_V_STATUS bit 0 is the VBlank flag. Reading TED_V_STATUS acknowledges that flag. TED_V_RASTER_STATUS is separate and remains pending until acknowledged with a write of 128.

This short polling example waits for scanline 100, changes the background to bright red, then acknowledges the compare:

```
10 TED ON
20 POKE &H000F0F20, &H18
30 POKE &H000F0F24, &H08
40 POKE &H000F0F30, &H76
50 POKE &H000F0F60, 100
60 POKE &H000F0F64, 0
70 IF (PEEK(&H000F0F68) AND 128)=0 THEN 70
80 POKE &H000F0F30, &H72
90 POKE &H000F0F68, 128
```

Raster interrupts are how the C16 and Plus/4 split the screen into multiple regions with different scroll, colour, or character-set choices. The same trick works here.

6.11 Copper-driven TED bands

The VideoChip copper can also write TED registers on selected scanlines. Use SETBASE with \$F0F20, then MOVE the TED register index. The register index is (register address - \$F0F20) / 4.

For example, TED_V_BORDER is \$F0F40, so its copper index is 8. This hand-packed list changes the TED border from red to green on scanline 80. The copper word format is documented in Chapter 4.

```
10 TED ON
20 POKE &H000F0F40, &H72
30 L=&H00003000
40 POKE L+0, &H8003C3C8 : REM SETBASE $F0F20
50 POKE L+4, &H00050000 : REM WAIT y=80, x=0
60 POKE L+8, &H40080000 : REM MOVE border index
70 POKE L+12, &H00000075 : REM green
80 POKE L+16, &HC0000000 : REM END
90 POKE &H000F0010, L : REM COPPER_PTR
100 POKE &H000F000C, 1 : REM COPPER_CTRL
```

You should see the top border in bright red and the lower border in bright green.

6.12 Setup order, side effects, and limits

The shortest TED program from machine language:

1. Write a non-zero value to TED_V_ENABLE (or use TED ON).

2. Set the picture mode in TED_V_CTRL1 and TED_V_CTRL2 (or use TED_MODE).
3. Choose where the video matrix and character set live with TED_V_VIDEO_BASE and TED_V_CHAR_BASE.
4. Fill the video matrix and colour RAM in the regions you chose.
5. To animate, poll TED_V_STATUS bit 0 for the vertical retrace, or arm a raster compare.

Important side effects and limits:

- Only the low byte of each TED video register is meaningful.
- TED_V_STATUS clears its VBlank bit when read.
- TED_V_RASTER_STATUS is sticky until bit 7 is written back as 1.
- TED_V_RASTER_LO and TED_V_RASTER_HI are read-only.
- TED_V_CTRL2 bit 5 is stored as RES, but it does not clear TED VRAM.
- Base registers select offsets inside the 16 KB TED VRAM window; an out-of-range selection falls back to the default base.
- Standard and extended-colour text use one-pixel horizontal detail. Multicolour text and multicolour bitmap use double-wide colour pixels.
- High-resolution bitmap mode uses only the two four-bit matrix nibbles for each cell. Multicolour bitmap adds the full colour-RAM byte as the fourth colour source.

The next chapter covers ANTIC and GTIA, the Atari 8-bit picture chips, which live on layer 13.